CSTE4103 · CLOUD COMPUTING

Lecture 1

# The Era of Cloud Computing

*Motivation for the Cloud · The Power Wall · Racks & Data Centers · Elastic Computing · Multi-tenant Clouds · IaaS / PaaS / SaaS / DaaS*

4th-Year Cybersecurity Engineering · Week 1 · Semester 7

Dept. of ICT — Polytechnic College for Engineering Specializations, MTU

TODAY'S ROADMAP

## Learning Objectives

1. Explain why computing moved from on-premise servers to the cloud
2. Describe the Power Wall and why it forced a shift to distributed systems
3. Understand elastic computing and multi-tenant architectures
4. Classify cloud services: IaaS, PaaS, SaaS, and DaaS
5. Recognize how each service model changes the security responsibility boundary

1. MOTIVATION

# Before the Cloud: The Bare-Metal Era

### A Traditional Data Room



*One organization → one server rack → fixed capacity*

Bought, owned, cooled, patched, and physically secured by the organization itself.

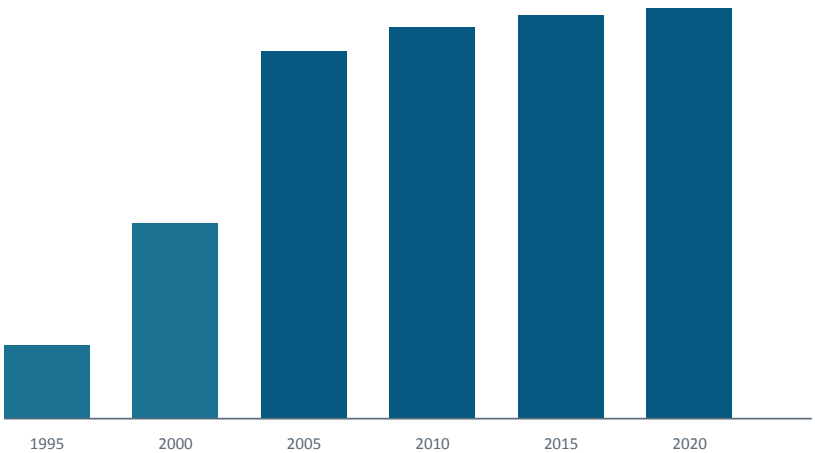
- High CAPEX: hardware purchased up front, before demand is known
- Capacity is fixed — over-provision (waste) or under-provision (outage)
- Every patch, backup, and physical control is the owner's job alone
- Scaling means buying, shipping, and racking new machines — weeks, not minutes



2. THE POWER WALL

# Why We Stopped Making Chips Faster

*Single-core clock speed (illustrative)*




Flattens after ~2005 — heat & power limits, not lack of demand

### The Power Wall

Faster clock speeds mean exponentially more heat and power draw. Around the mid-2000s, chipmakers hit a physical ceiling: a single processor could no longer get meaningfully faster without melting itself.

The industry's answer: stop chasing raw speed. Add more cores, and more machines, and spread the work across them — parallel and distributed computing.

*This is the physical reason cloud data centers are built from thousands of ordinary machines rather than one extraordinary one.*

3. SCALING OUT

# Racks of Server Computers

The horizontal answer to the Power Wall: instead of one powerful machine, use many ordinary ones, organized for scale.

**Server**

A single physical machine — CPU, RAM, disk, NIC

→

**Rack**

10–40+ servers stacked in one cabinet, sharing power & networking

→

**Aisle / Pod**

Rows of racks grouped for airflow, power, and network design

→

**Data Center**

Thousands of racks — the physical body of "the cloud"

*Full data-center design (cooling, North-South/East-West traffic, fat-tree & leaf-spine networks) is covered in Week 3 — today, just the concept.*



4. ELASTICITY

# Elastic Computing

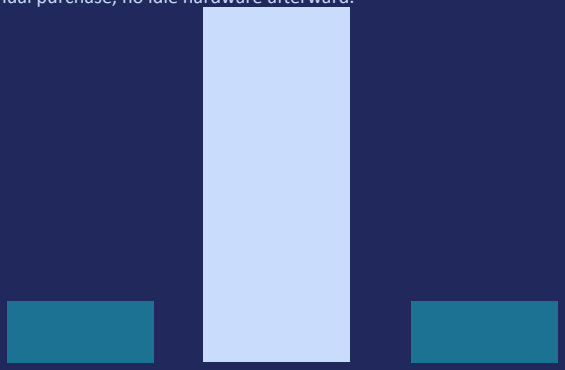
## Definition

The ability to automatically acquire more computing resources when demand rises, and release them when demand falls — paying only for what is actually used.

*Contrast with the bare-metal era: capacity there was fixed at purchase time, months before it was needed.*

**Example — Exam-Result Portal Traffic**

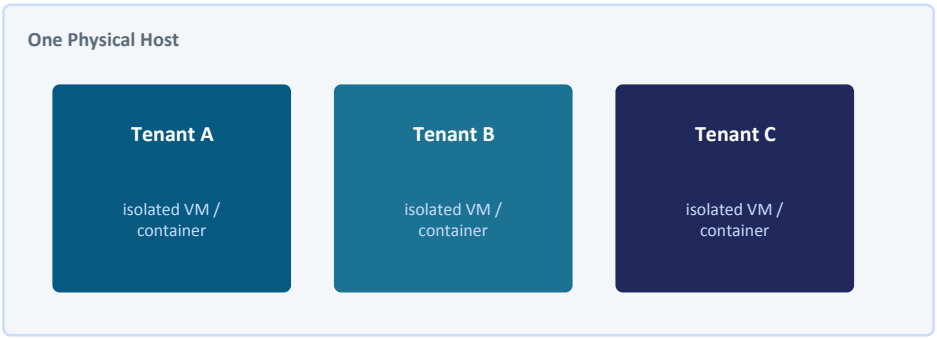
Servers scale up automatically for the traffic spike, then scale back down — no manual purchase, no idle hardware afterward.



5. SHARED INFRASTRUCTURE

# Multi-tenant Clouds

Definition: many independent customers ("tenants") share the same underlying physical hardware, while each tenant's data and workloads remain logically isolated.




## Security Angle

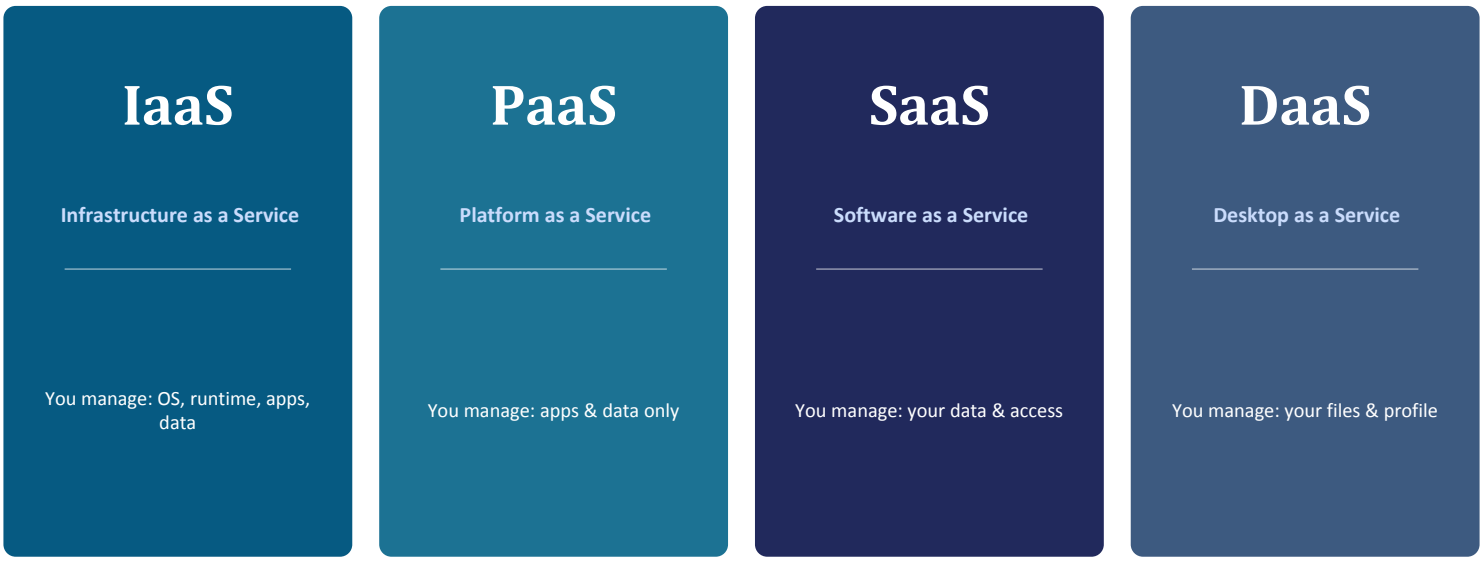
Shared hardware means a new attack surface: hypervisor escapes, side-channel attacks (e.g. cache timing), and noisy-neighbor resource abuse — none of which exist in a single-tenant, bare-metal setup.

*Isolation is enforced in software (hypervisor / container runtime) — its correctness is now a core security control.*



6. SERVICE MODELS

# IaaS · PaaS · SaaS · DaaS



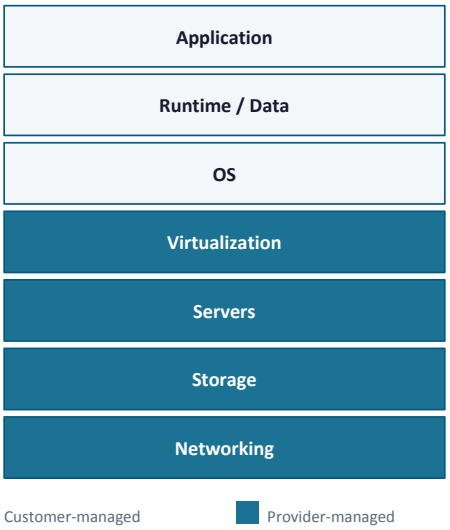

*Left → Right: the provider takes on more of the management burden — and more of the security responsibility along with it.*

IAAS

# Infrastructure as a Service

The provider supplies raw compute, storage, and networking — virtualized hardware. You install and manage everything above it.

- Examples: Amazon EC2, Azure Virtual Machines, Google Compute Engine
- You choose the OS, install your own stack, configure networking rules
- Maximum control and flexibility — maximum operational responsibility
- Typical users: system administrators, DevOps teams, infrastructure engineers



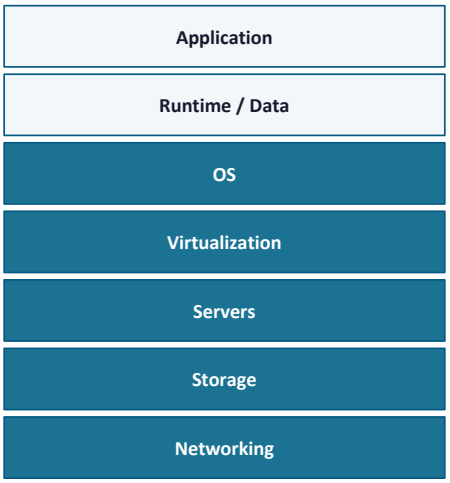



PAAS

# Platform as a Service

The provider manages the OS, runtime, and middleware. You deploy code and data — the platform handles patching, scaling, and availability.

- Examples: Azure App Service, Google App Engine, Heroku
- Developers push code; the platform builds, deploys, and scales it
- No OS patching, no server maintenance — faster development cycles
- Typical users: application developers, small engineering teams

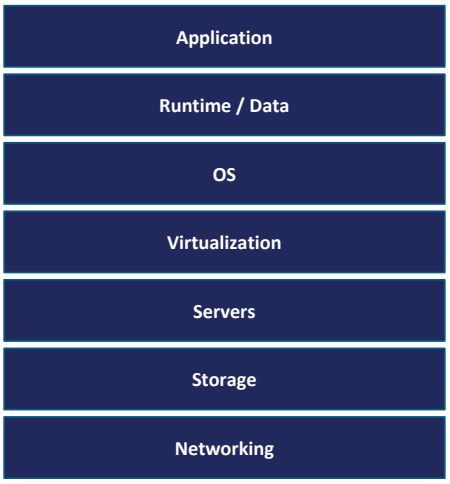


SAAS

# Software as a Service

The provider runs the entire application. You simply use it — typically through a browser — and manage only your own data and user access.

- Examples: Google Workspace, Microsoft 365, Salesforce, Zoom
- Nothing to install, patch, or scale — the vendor owns the full stack
- Your responsibility narrows to: account security, access control, data you upload
- Typical users: everyone — end users, administrators, entire organizations




*Customer still owns: identity & access management, data classification, and end-user device security.*



DAAS

# Desktop as a Service

A complete virtual desktop — OS, applications, and user profile — streamed from the cloud to any device.

### Examples

Amazon WorkSpaces, Windows 365, Citrix DaaS

### Why organizations use it

Consistent, centrally-managed desktops for remote or contract staff; sensitive data never leaves the data center — the endpoint only displays pixels.

*This last point matters for security: a lost or compromised laptop does not mean lost data.*

### Where DaaS sits

DaaS is really SaaS applied to the desktop itself: the provider manages the full OS + app stack, and the customer manages only data, identity, and what the user does inside the session.

Typical users: remote workforces, BPO/call centers, students needing access to specialized lab software from any device.

BRINGING IT TOGETHER

# Why This Matters for a Security Engineer

| | |
|---|---|
| **Shared Responsibility** | Every cloud service has a line: what the provider secures, and what you secure. That line moves with the service model — know exactly where it sits before you design a control around it. |
| **New Attack Surface** | Multi-tenancy, hypervisors, and APIs are attack surfaces that simply did not exist in the bare-metal era. |
| **Elasticity Cuts Both Ways** | Auto-scaling defends against load spikes — but a misconfigured resource can also scale an attacker's cost-abuse or cryptomining attack. |



WRAP-UP

# Summary & Next Week

*Today, in one sentence:*

The Power Wall forced computing to scale horizontally, giving rise to elastic, multi-tenant clouds offered at four levels of abstraction — IaaS, PaaS, SaaS, DaaS — each shifting the security boundary.

---

## Next Week — Week 2

Types of Clouds and Their Providers: Private vs. Public Cloud, Provider Lock-in, Hybrid Cloud, Multi-Cloud, and Hyperscale providers.

## Discussion Questions

- If a bank moves its core banking app from bare-metal to SaaS, what security controls does it lose direct control over?
- Where would you place a national ID database on the IaaS–DaaS spectrum, and why?